

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                  |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
|             | <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                  |
|             | <b>Supplementary End Semester Examination – Summer 2022</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                  |
|             | <b>Course: B. Tech.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Branch:</b>                   |
|             | <b>Subject Code &amp; Name: BTES105</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Semester: II</b>              |
|             | <b>Max Marks: 60</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Date:</b>                     |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Duration: 3 Hr.</b>           |
|             | <b>Instructions to the Students:</b> <ol style="list-style-type: none"> <li>1. All the questions are compulsory.</li> <li>2. Each question carries 12 marks.</li> <li>3. Illustrate your answers with neat sketches, diagram etc., wherever necessary</li> <li>4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.</li> <li>5. Use of non-programmable scientific calculators is allowed.</li> <li>6. Assume suitable data wherever necessary and mention it clearly.</li> </ol> |                                  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | (Level/CO)                       |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks                            |
| <b>Q. 1</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                  |
| A)          | What are the biggest problems with using fossil fuels for energy?<br>Compare the Diesel based power plant with the Hydro energy power plant in respect of site requirement, environmental effect, initial cost, fuel transportation, reliability, operating cost, simplicity, and cleanliness.                                                                                                                                                                                                                                                                                           | Remember/CO1<br>Understand/CO1   |
| B)          | Explain the function of following components used in Gas Turbine power plant:<br>i) combustion chamber ii) Compressor iii) Turbine                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Remember/CO1                     |
| C)          | Explain the function of economizer and air preheater in Coal based power plant.<br>Write any three advantages and disadvantages of the coal based power plant.                                                                                                                                                                                                                                                                                                                                                                                                                           | Remember/CO1<br>Remember/CO1     |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                  |
| <b>Q.2</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                  |
| A)          | Explain the working principle of solar flat plate collector with a simple sketch.<br>Write at least three applications of solar flat plate collector.                                                                                                                                                                                                                                                                                                                                                                                                                                    | Understand/CO2<br>Remember/CO2   |
| B)          | Explain the working principle of MHD Generator with a neat diagram.<br>What are its advantages and limitations?                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Understand/CO2<br>Remember/CO2   |
| C)          | Write any four parameters that affect the efficiency of wind mill.<br>Draw a sketch of HAWT and label the important parts on it.                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Remember/CO2<br>Understand/CO2   |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                  |
| <b>Q. 3</b> | <b>Solve the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                  |
| A)          | Explain the meaning of 5 star rating on the home appliances.<br>How the energy conservation practices can be employed while in transportation?                                                                                                                                                                                                                                                                                                                                                                                                                                           | Understand/CO3<br>Understand/CO3 |
| B)          | Define Energy Conservation.<br>Write the various measures that makes energy saving while using the air conditioner at home.                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Remember/CO3<br>Understand/CO3   |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                  |
| <b>Q.4</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                  |
| A)          | Differentiate between primary and secondary air pollutants.<br>Write any two major accidents happened due to air pollution in the world and explain their causes.                                                                                                                                                                                                                                                                                                                                                                                                                        | Understand/CO4<br>Remember/CO4   |

|             |                                                                                                                                      |                                |                      |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------|
| <b>B)</b>   | Which is the biggest source of air pollution in the world today?<br>How it can be prevented?                                         | Remember/CO4<br>Understand/CO4 | <b>2</b><br><b>4</b> |
| <b>C)</b>   | What is the role of an individual in prevention of pollution?<br>Explain the health effects of air pollution on the living organism. | Understand/CO4<br>Remember/CO4 | <b>2</b><br><b>4</b> |
|             |                                                                                                                                      |                                |                      |
| <b>Q. 5</b> | <b>Solve Any Four of the following.</b>                                                                                              |                                |                      |
| <b>A)</b>   | What are the sources of soil pollution and how these problems can be solved?                                                         | Remember/CO5                   | <b>3</b>             |
| <b>B)</b>   | Explain the effect of water pollution on the human health.                                                                           | Remember/CO5                   | <b>3</b>             |
| <b>C)</b>   | What are 3 issues of polluting the ocean?<br>How does marine pollution affect climate change?                                        | Understand/CO5                 | <b>3</b>             |
| <b>D)</b>   | Give the difference between BOD and COD in terms of decomposition and measurement limit.                                             | Remember/CO5                   | <b>3</b>             |
| <b>E)</b>   | How can we protect us from the noise pollution?                                                                                      | Understand/CO5                 | <b>3</b>             |
| <b>F)</b>   | What is radioactive pollution? How one can prevent it?                                                                               | Remember/CO5                   | <b>3</b>             |
|             | <b>*** End ***</b>                                                                                                                   |                                |                      |

**The grid and the borders of the table will be hidden before final printing.**